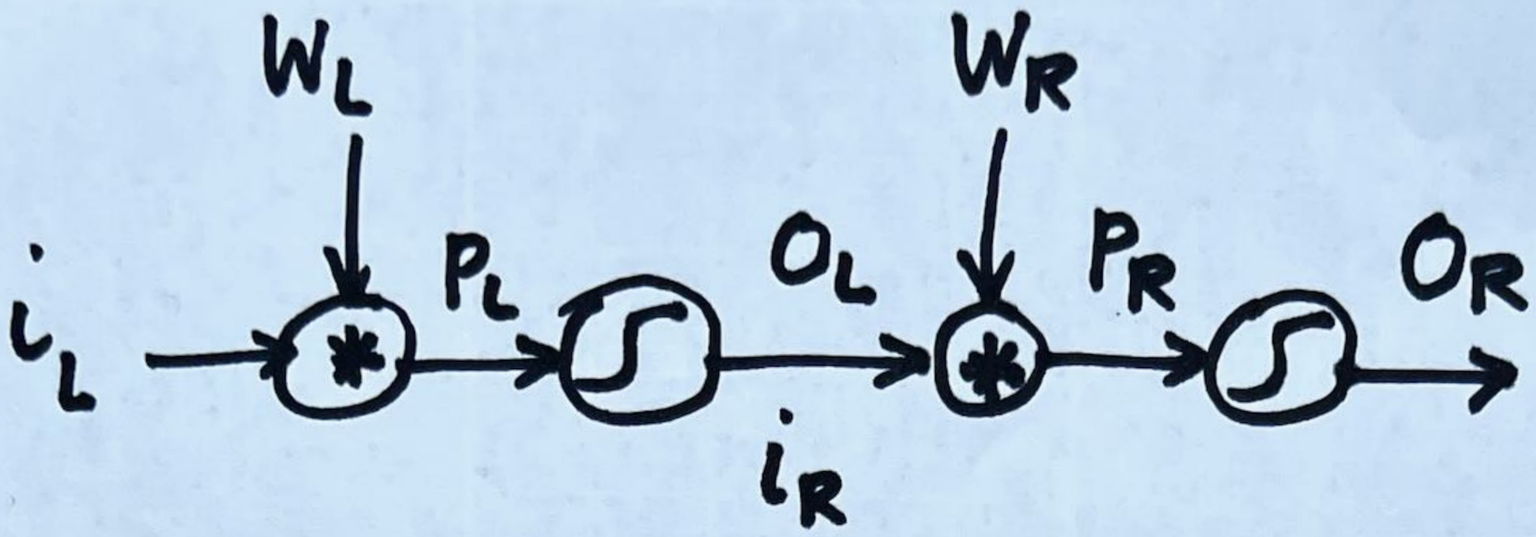
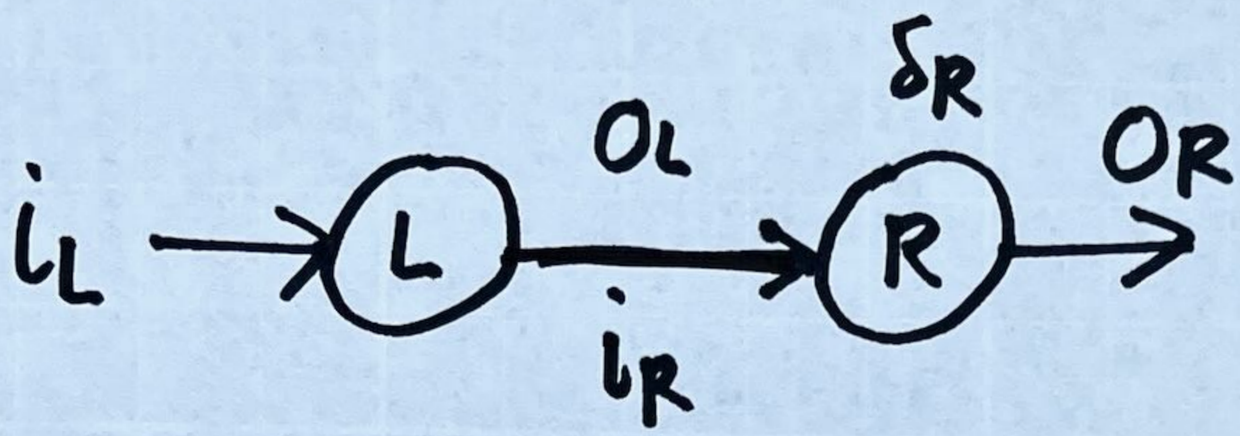
Derivation of Backprop Formulas





$$y = \frac{1}{1+e^{-x}} \quad \text{Sigmoid}$$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{1+e^{-x}}\right)$$

$$= \frac{d}{dx}\left((1+e^{-x})^{-1}\right)$$

let $u = 1+e^{-x}$

$u^n = (1+e^{-x})^{-1}$

$$= (-1) * (1+e^{-x})^{-1-1} \frac{d}{dx}(1+e^{-x})$$

$$= (-1) * (1+e^{-x})^{-2} * \left(\frac{d(1)}{dx}^{0} + \frac{d(e^{-x})}{dx}\right)$$

$$= (-1) * (1+e^{-x})^{-2} * \left(e^{-x} * \frac{d(-x)}{dx}\right)$$

$$= (-1) * (1+e^{-x})^{-2} * (e^{-x} * (-1))$$

$$= \frac{e^{-x}}{(1+e^{-x})^2} = \frac{e^{-x}}{(1+e^{-x})(1+e^{-x})}$$

$$\frac{d\,u^n}{dx} = n\,u^{n-1}\frac{du}{dx}$$

$$\frac{d\,e^u}{dx} = e^u \frac{du}{dx}$$

$$= \frac{1 + e^{-x} - 1}{(1+e^{-x})(1+e^{-x})}$$

$$= \left(\frac{1}{1+e^{-x}}\right) * \left(\frac{1+e^{-x}}{1+e^{-x}} - \frac{1}{1+e^{-x}}\right)$$

$$\boxed{\frac{dy}{dx} = y * (1 - y)}$$

Derivative of P with respect to the actual output of the network

let d - desired output

o - actual output

$$P = \left(-\frac{1}{2}\right)(d-o)^2$$

$$\frac{dP}{do}$$

let $u = (d-o)$

$$\frac{dP}{do} = \left(-\frac{1}{2}\right)(2)(d-o)^{2-1}\frac{d(d-o)}{do}$$

$$= (-1)(d-o)(-1)$$

$$\boxed{\frac{dP}{do} = (d-o)}$$

$$\frac{dv^n}{dx} = nv^{n-1}\frac{dv}{dx}$$

$$\Delta W_R \propto \frac{\partial P}{\partial W_R}$$

$$\frac{\partial P}{\partial W_R} = \frac{\partial P}{\partial O_R} \times \frac{\partial O_R}{\partial W_R}$$

$$(d - O_R) \times \frac{\partial O_R}{\partial P_R} \times \frac{\partial P_R}{\partial W_R}$$

$$O_R(1 - O_R)$$

$$\frac{\partial (W_R \times O_L)}{\partial W_R}$$

$$\frac{\partial P}{\partial W_R} = [d - O_R] \times [O_R(1 - O_R)] \times [O_L]$$

$$\frac{\partial P}{\partial W_R} = \boxed{[d - O_R] \times [O_R(1 - O_R)] \times i_R}$$

$$\Delta W_L \propto \frac{\partial P}{\partial W_L}$$

$$\frac{\partial P}{\partial W_L} = \frac{\partial P}{\partial O_R} \times \frac{\partial O_R}{\partial W_L}$$

$$(d - O_R) \times \frac{\partial O_R}{\partial P_R} \times \frac{\partial P_R}{\partial W_L}$$

$$O_R(1 - O_R)$$

$$\frac{\partial P_R}{\partial O_L} \times \frac{\partial O_L}{\partial W_L}$$

$$\frac{\partial (W_R \times O_L)}{\partial O_L} \times \frac{\partial O_L}{\partial P_L} \times \frac{\partial P_L}{\partial W_L}$$

$$\frac{\partial (W_L \times i_L)}{\partial W_L}$$

$$= [d - O_R] \times [O_R(1 - O_R)] \times [W_R] \times [O_L(1 - O_L)] \times [i_L]$$

$$\frac{\partial P}{\partial W_R} = [d - O_R] \times [O_R(1 - O_R)] \times [i_R]$$

$$\frac{\partial P}{\partial W_L} = [d - O_R] \times [O_R(1 - O_R)] \times [W_R] \times [O_L(1 - O_L)] \times [i_L]$$

To simplify,

Let $\delta_R = (d - O_R) \times [O_R(1 - O_R)]$

$$\delta_L = \delta_R \times W_R \times [O_L(1 - O_L)]$$

$$\frac{\partial P}{\partial W_R} = \delta_R \times i_R$$

$$\frac{\partial P}{\partial W_L} = \delta_L \times i_L$$

$$W_R = W_R + n\frac{\partial P}{\partial W_R}$$

$$W_L = W_L + n\frac{\partial P}{\partial W_L}$$

Final output layer (output neurons)

$$W_F = W_F + n\frac{\partial P}{\partial W_F}$$

For all other neurons in the other layer

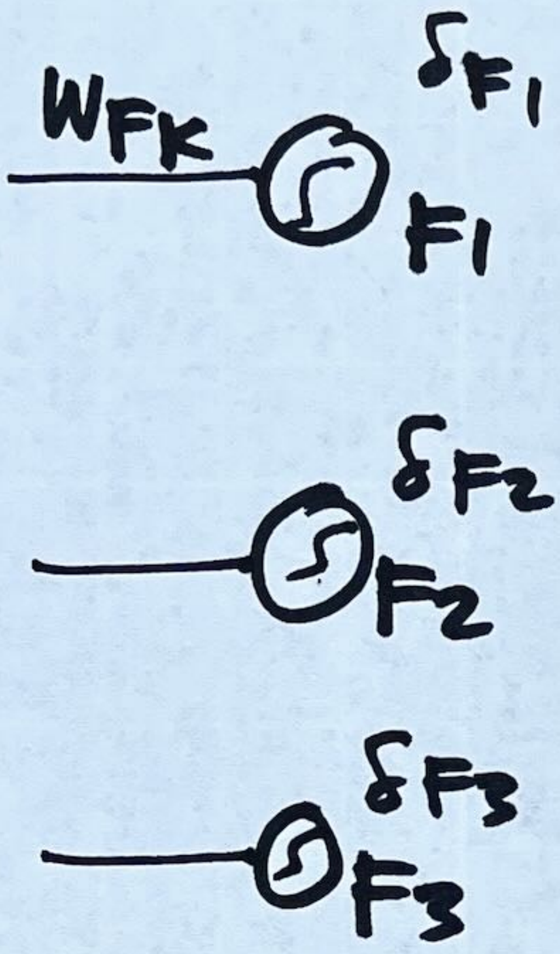
$$W_L = W_L + n\delta_L i_L$$

$$\delta_L = \delta_R \times W_R \times O_L(1-O_L)$$

Generalisation

For networks with multiple neurons par layer

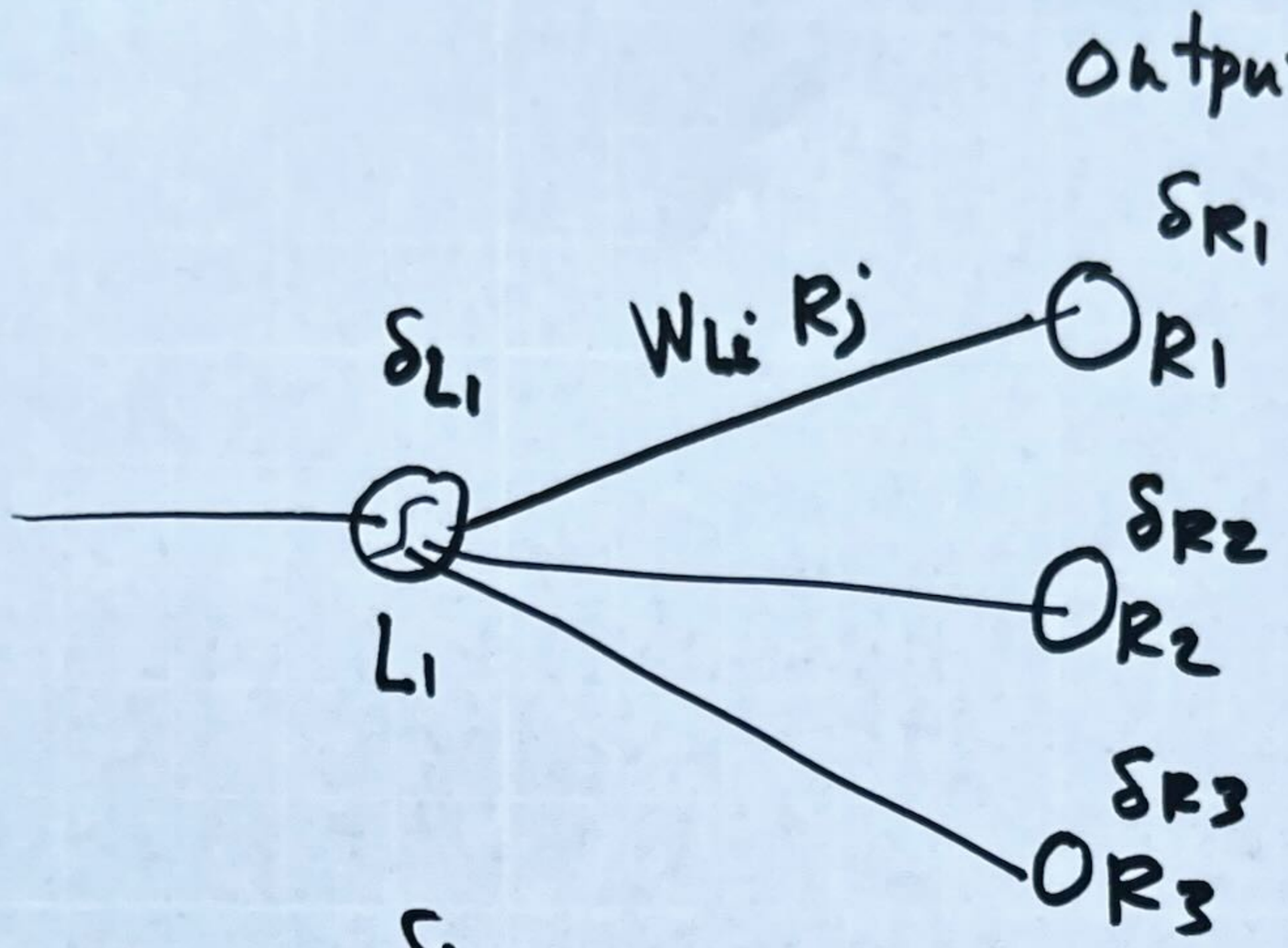
Final layer neurons :



$$W_{Fk} = W_{FK} + n \delta_{Fk} i_{Fk}$$

$$\delta_{Fk} = (d_k - o_{Fk}) \times [o_{Fk}(1 - o_{Fk})]$$

Other neurons (not in the output layer)



(right-hand side)

no. of neurons connected to Li

$$\delta_{Li} = \left[\sum_{j=1}^{m} \left(\delta_{Rj} \times W_{Li\,Rj} \right) \right] \times O_{Li}(1 - O_{Li})$$